

Comparative Analysis of Random Forest vs Isolation Forest Algorithms in Detecting Timestamp Manipulation on NTFS Using the \$LogFile and \$UsnJrnl Artifacts

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Abstract— Timestamp manipulation, or timestomping, is a critical anti-forensic technique used by cybercriminals to obscure the timeline of malicious activity by altering file metadata, specifically the modified, accessed, changed, and created or birth timestamps in the New Technology File System. Existing rule-based detection tools often struggle to differentiate between malicious timestomping and benign system activities, leading to high false positives that compromise the efficiency of forensic investigation. This paper addresses this gap by reframing timestomping detection as an anomaly detection problem with a severe class imbalance. An artificial intelligence-driven methodology was proposed and evaluated by using and comparing the supervised Random Forest classifier and the unsupervised Isolation Forest model for anomaly detection, trained on a controlled, labeled dataset derived from the feature-engineered correlation of the New Technology File System’s journaling artifacts, the Log File and the Update Sequence Number Change Journal. The final model classifies file system events with a measurable confidence score. Between the two models evaluated, the Random Forest classifier has demonstrated superior performance, significantly reducing the false positive rate compared to the Isolation Forest model. This validation of a robust data-driven approach for timestomping detection, which is also packaged as a Python Data Source Ingest Module plugin prototype for Autopsy that automates the entire detection pipeline, provides digital forensic investigators with a confidence-scored result to identify file manipulation, thereby enhancing the reliability and efficiency of artifact anomaly detection in the digital forensic workflow.

Keywords— *Timestamp manipulation, New Technology File System (NTFS) artifacts analysis, Digital Forensics, anomaly detection, Machine Learning.*

I. INTRODUCTION

Cybercrime continues to escalate globally, exploiting vulnerabilities in digital systems to conceal malicious activity.

One of the most common strategies employed by cybercriminals is anti-forensics, which involves deliberately manipulating digital evidence to hinder investigations [1]. Moreover, they are increasingly utilizing timestamp manipulation, also known as “timestomping,” to change the date and time information associated with files, so it creates illusion of no suspicious activity. This activity causes challenges for digital forensic investigators because they need to determine the timeline of events, including when the files are created, accessed, modified, or deleted. Timestomping creates a major threat that weakens the strength of forensic evidence and integrity of the investigation. Given the growing reliance on digital evidence in criminal cases and organizational investigations, accurate timeline reconstruction has become central to modern digital forensics.

Timeline reconstruction primarily relies on system artifacts, such as logs, registry data, and file system metadata, the descriptive information stored about files. In Windows, the New Technology File System (NTFS) stores multiple timestamp attributes—Modified, Accessed, Changed, and Birth (MACB)—which reflects how files have been created, used, or altered [2]. Discrepancies observed in these attributes can reveal if tampering occurred and can help in reconstructing sequence of file system events that occurred [3].

The NTFS transaction log, which is the \$LogFile is frequently used for its capabilities to record file system transactional changes, including file creation, deletion, and data change. Complementing this, the \$UsnJrnl or Update Sequence Number Journal, acts as the supporting artifact because it keeps a log of all file activities, such as file creation, deletion, renaming, and modification [4], [2]. This NTFS journal artifact provides the historical record of file changes and shows what change occurred. These NTFS artifacts are important in identifying user actions, linking digital behavior to specific timeframes, and validating other sources of evidence.

Forensic studies show that both legitimate user operations and malicious timestomping tools can generate irregular and inconsistent timestamp patterns, making it difficult to distinguish between benign and malicious behavior [5]. As a result, false positives remain a persistent challenge in timestamp analysis. Effective detection of anti-forensic activity like timestomping cannot be done by looking at a single artifact. Doing so may lead to false positives and incomplete findings. Checking multiple NTFS artifacts improves timestamp manipulation detection accuracy and strengthens the credibility of forensic conclusions. To address this issue, this study proposes a data-driven methodology comparing Random Forest and Isolation Forest algorithms in detecting timestamp manipulation on NTFS using \$LogFile and \$UsnJrnl artifacts. By analyzing metadata in both NTFS artifacts, the comparative analysis highlights differences in timestomping detection accuracy and false positives rates. This provides insight into selecting the most effective algorithm in detecting timestamp manipulations within Windows NTFS. With automated detection, AI-driven models will be able to ensure greater efficiency and higher accuracy in timeline reconstruction.

Ultimately, the primary objective of this study is to integrate digital forensic reasoning with machine learning to create a Python Data Source Ingest Module for Autopsy, a widely used digital forensics platform. This plugin will aid investigators in automating timestamp analysis and providing results with confidence scores and anomaly detection results.

A. Significance of the Study

The study contributes to cybersecurity and digital forensics by integrating machine learning in NTFS artifact analysis, specifically \$LogFile and \$UsnJrnl, in automating timestamp detection. This study bridges traditional rule-based methods, and modern AI-driven approaches, enabling more data-driven and adaptive forensic analysis.

The results benefit law enforcement agencies (PNP-ACG) and digital forensic practitioners by improving timestamp detection accuracy and reliability of digital evidence. Furthermore, the study will provide the academic and research community with a model and framework that may serve as their foundation for future AI-driven forensic tools.

B. Objectives of the Study

The general objective of this study is to develop and evaluate a confidence-based machine learning framework for timestomping detection in NTFS artifacts, specifically the \$LogFile and \$UsnJrnl, and implement it as a Python Data Source Ingest Module plugin prototype for Autopsy. The specific objectives of the study are the following: (1) to compare the performance of two different trained machine learning classification models for detecting timestomping within a feature-engineered dataset derived from correlated \$LogFile and \$UsnJrnl metadata, (2) to refine and validate the superior machine learning model to reliably classify malicious and benign file system events with a measurable confidence score, and (3) to implement the validated

classifier as a Python Data Source Ingest Module plugin prototype for Autopsy that automates the entire timestomping detection pipeline for practical use of digital forensic investigators in their forensic workflows.

II. REVIEW OF RELATED STUDIES

A. Synthesis of Related Studies

The reviewed studies collectively emphasize the increasing complexity of detecting timestamp manipulation within NTFS. Study [2] shows that inconsistencies across \$MFT, \$LogFile, and \$UsnJrnl can indicate timestomping, yet rule-based approaches still produce high false positives due to benign system activities that resemble manipulation. Work in [5] further demonstrates how NTFS timestamps behave differently across operating system versions and software workflows, making isolated timestamp analysis unreliable.

Research in [6] expands this by showing that NTFS timestamps can correspond to multiple valid file histories, highlighting the need for multi-artifact correlation instead of direct linear interpretation. Broader anti-forensic literature also underscores the sophistication of obfuscation methods. Reference [7] classifies timestomping within wider anti-forensic strategies that distort or conceal evidence, while [8] shows how timestamp tampering undermines event reconstruction and necessitates more advanced analytical models.

Recent studies on evasion techniques [9] and data trace reliability [10] further illustrate how attackers exploit metadata ambiguity and how forensic tools must adapt to maintain evidentiary integrity. Supporting this, [11] highlights the forensic relevance of timestamp-related contact artifacts, whereas [12] identifies the increasing importance of AI-driven methodologies in modern forensic workflows.

Overall, the literature identifies three key gaps: (1) NTFS timestamps are inherently ambiguous and sensitive to benign variation, (2) modern anti-forensic techniques reduce the reliability of traditional rule-based detection, and (3) machine-learning-based, multi-artifact analysis remains underexplored. These gaps justify developing a data-driven, AI-enabled approach using correlated \$LogFile and \$UsnJrnl artifacts for more reliable detection of timestomping.

III. METHODOLOGY

The methodological framework for this study is a comparative data-driven approach designed to address the severe class imbalance inherent in timestomping detection. It is divided into three primary phases: data preparation, feature engineering, and the design of the comparative machine learning pipeline.

A. Data Acquisition & Preparation

The foundation of the research is a forensically relevant, labeled dataset constructed from 12 distinct forensic cases provided by Junghoon Oh, each containing known timestomping activity.

The initial stage involved Parsing and Extraction, where raw \$LogFile and \$UsnJrnl artifacts were systematically processed using the NTFS Log Tracker tool to convert them into a structured Comma-Separated Value (CSV) formats. This was followed by Labelling and Cleaning to refine the data. Data cleaning dropped columns and rows lacking forensic value, thereby reducing noise. Data labeling was executed by cross-referencing records with the tool's Suspicious-Behavior-Detected output, consistently assigning the anomaly class to timestamped events.

A critical step was Cross-Artifact Integration, where the individual sets of \$LogFile and \$UsnJrnl records were consolidated into unified case files. A necessary outer join strategy was performed on file path and timestamps fields to merge the complementary but non-overlapping information from the two artifact types. This ensures that valuable traces left in either journal artifact are preserved and correlated.

Finally, the Master Timeline was created by vertically concatenating all unified case files into a single chronologically ordered dataset. This aggregated dataset is designated for a case-based stratified split to prevent data leakage and ensure the model is robustly tested on entirely unseen forensic scenarios, simulating real-world application.

B. Feature Engineering

Following the data collection and preprocessing phase, an extensive feature engineering process was conducted to transform the raw forensic event data into a machine learning-ready format. The main goal was to develop features capable of detecting the nuanced patterns and characteristics traces left by timestamp manipulation tools. The developed features were logically grouped into five core categories:

1) *Temporal Features*. These features were created to capture time-based patterns that deviate from normal system activity or may suggest manipulation. This includes analyzing time components, deriving behavioral flags, calculating time delta relations, and identifying activity patterns, based on the rationale that timestomping often occurs during off-hours or results in event clustering distinct from typical system behavior.

2) *Timestomp Anomaly Features*. This category focuses on detecting logical inconsistencies in the MAC (Modified-Accessed-Created) timestamps. Features include flagging files where the creation timestamp is later than the modification timestamp, identifying suspiciously identical MAC timestamps, and capturing abnormal temporal gaps. Such anomalies violate fundamental filesystem semantics and serve as strong indicators of manipulation.

3) *Path Characteristics*. These features analyze file location and naming patterns, including path analysis, location

flags, and file metadata. This is relevant as anti-forensic tools often target specific executable files or known sensitive paths.

4) *Event Pattern Features*. These encode forensic event sequences found in both the \$LogFile and \$UsnJrnl, analyzing the rarity of displayed events and their sequential patterns. This leverages the principle that timestomping tools generate characteristic event sequences in journal files that differ from normal system operations.

5) *Cross-Artifact Features*. As a key innovation of this study, this category comprises features derived from the combined \$LogFile and \$UsnJrnl data, including merge indicators, artifact presence flags, file attributes, and label source tracking, emphasizing the correlation of the two complementary journals.

C. Machine Learning Pipeline Design

The selection of machine learning models is driven by comparative analysis of existing timestamp manipulation detection approaches. It is important to note that a direct numerical baseline for a comparable machine learning model that leverages both \$LogFile and \$UsnJrnl is not yet available, establishing this research as a foundational benchmark.

Two complementary machine learning approaches are employed to evaluate different detection methods: supervised learning with Random Forest and unsupervised anomaly detection with Isolation Forest.

1) *Random Forest Classifier (Supervised Approach)*. A Random Forest Classifier was selected as the primary supervised learning model. This ensemble decision tree model was chosen over simpler linear models because it inherently captures complex, non-linear patterns, effectively handles the severely imbalanced nature of the forensic dataset, and provides built-in mechanisms against overfitting.

The model's training and validation phase utilizes a case-based stratified split of the master timeline, assigning entire forensic cases to either the training or testing set. This prevents data leakage and provides a realistic measure of performance on completely unseen cases. The training strategy for the imbalanced data employs a two-part approach: the Synthetic Minority Over-sampling Technique (SMOTE) is used to create a controlled number of artificial timestomped samples (1:1000 ratio, creating only 338 synthetic samples), and an added penalty is implemented via moderate class weighting (5:1 ratio) to ensure the model appropriately penalizes the misclassification of timestomped files while maintaining precision.

2) *Isolation Forest (Unsupervised Anomaly Detection)*. To complement the supervised approach, an Isolation Forest model is implemented to evaluate whether timestomped events exhibit anomalous patterns detectable through unsupervised learning.

Isolation Forest operates on the principle that anomalies are few and different, requiring fewer random partitions to isolate outliers from normal samples. Unlike Random Forest, this approach does not require synthetic oversampling, as it directly identifies anomalies based on isolation depth in randomly constructed trees. The contamination parameter is calibrated to the observed timestamped event rate in the training data.

Feature scaling via StandardScaler is applied prior to training, as Isolation Forest is sensitive to feature magnitudes. The model assigns anomaly scores to each event, where lower scores indicate higher anomaly likelihood. This unsupervised paradigm tests the hypothesis that timestamp manipulation creates forensic artifacts sufficiently distinct from benign file activity to be detected without explicit labeling

3) *Model Evaluation Framework.* Both models' performance is assessed through a multi-faceted evaluation process using metrics specifically chosen for extreme class imbalance. The primary evaluation metrics include Precision (to minimize false positives), Recall (to maximize detection), and the F1-Score (to provide a single consolidated measure of triage performance). Furthermore, the Area Under the Precision-Recall Curve (AUC-PR) is selected as the key summary metric, as it offers a more informative overview for severely imbalance data than the traditional AUC-ROC.

For Random Forest, a feature importance analysis identifies the most predictive forensic indicators, ranked by their contribution to classification accuracy. For Isolation Forest, anomaly score distributions are analyzed to assess the separability between benign and timestamped events. A direct comparative analysis evaluates both models on identical test sets to determine the optimal detection approach for forensic timestamping detection.

D. Prototype Integration

The prototype will be developed as a Python Data Source Ingest Module for the Autopsy digital forensics platform. The workflow will begin within the Autopsy interface, where it will allow an investigator to add a data source, such as a disk image, to their created case. This component will then automatically identify and export the \$LogFile and \$UsnJrnl raw files to an Autopsy case-specific folder.

Once these artifacts are exported, the core of the timestamp detection framework, an external Python 3 processing with a trained machine learning model, will be automatically invoked. This external process will serve as the analytical core of the prototype, first parsing the raw NTFS artifacts into a structured format. After this, it will merge the data into a master timeline. With the Random Forest model embedded, the engineered feature set for each file system event will be passed to the external processing layer. Finally, this engineered data will be run through the trained model to generate the final predictions.

The last process will give these results back to the investigator. It will read the output file generated by the external process layer and will display the findings, including the confidence score for each event, within Autopsy's results viewer interface. This final step will provide a clear and actionable report, allowing the investigator to quickly identify and prioritize the most suspicious file system activities.

IV. PRELIMINARY RESULTS & DISCUSSION

This section presents the quantitative comparison between the primary supervised machine learning approach (Random Forest) and an unsupervised baseline (Isolation Forest). The results validate the effectiveness of the supervised framework in the context of extreme class imbalance and provide a foundational benchmark for forensic timestamping detection.

A. Preliminary Results

1) *Model Performance Comparison.* Both models trained and tested on the same dataset, representing a severe class imbalance ratio. The performance on the held-out test set is summarized in the Table 1 below.

TABLE I. COMPARATIVE MODEL PERFORMANCE METRICS

Metric	Random Forest	Isolation Forest	Difference
Precision	0.427 (42.7%)	0.004 (0.4%)	-42.3%
Recall	0.657 (65.7%)	0.010 (1.0%)	-64.7%
F1-Score	0.517	0.006	-0.511
AUC-ROC	0.999	0.947	-0.052
AUC-PR	0.348	0.005	-0.343
False Positive Rate (FPR)	0.030%	0.082%	+0.052%

The results establish a substantial performance gap between the two approaches, confirming the superior effectiveness of the supervised learning approach for this problem. The Random Forest classifier demonstrated an F1-Score eighty-nine times greater than that of the Isolation Forest.

2) *Random Forest Performance (Supervised Approach).* The Random Forest model, configured with a minimal SMOTE ratio and a moderate class weighting penalty, achieved balanced and forensically valuable performance on the test set. The Precision of 42.7% indicated that out of every flagged file, the said result was genuine timestamped detections minimizing wasted investigator effort. A recall of 65.7% presents the successful rate of actual timestamping files, ensuring the capture most of malicious activity.

3) *Feature Importance Analysis.* The Random Forest model provided crucial interpretability via feature importance analysis, identifying the forensic indicators most predictive of manipulation. The top features reveal that the model relies heavily on pattern recognition within the journals rather than simple timestamp checks.

TABLE II. TOP FIVE RANDOM FOREST FEATURE IMPORTANCE

Rank	Feature	Importance	Category
1	usn_event_encoded	10.72%	Event Patterns
2	lf_event_encoded	9.99%	Event Patterns
3	is_temp_path	9.59%	Path Features
4	events_per_minute	8.80%	Temporal Features
5	has_both_artifacts	7.40%	Cross-Artifact

This finding is the critical differentiator from the unsupervised approach. Traditional timestamp anomaly flags, which rely on individual feature deviations, contributed only marginally to the model's predictive power. The Isolation Forest, which fundamentally relies on finding extreme deviations, fails because timestomping manipulation does not create an extreme deviation in individual features; rather, it creates a subtle, learned pattern of inconsistency across multiple journal entries that only the supervised Random Forest can detect. This confirms the value of correlating data from both the \$LogFile and \$UsnJrnl as a necessary innovation.

B. Discussion

1) *Random Forest Superiority.* The profound difference in performance, particularly the sixty-four point seven percent higher Recall achieved by the Random Forest classifier, validates the hypothesis that supervised learning with explicit labeling is essential for timestamp manipulation detection. The findings suggest that the artifacts left by timestomping tools (like NTimeStomp and SetMACE) do not consistently present as extreme statistical outliers, which is the foundational assumption of anomaly detection methods like Isolation Forest. Instead, the manipulation creates subtle inconsistencies, such as cross-artifact mismatches and event sequence disruptions, that require contextual pattern learning.

The high importance of features like usn_event_encoded and the cross-artifact features demonstrates that the model is learning the relationship between events, a capability that unsupervised methods lack. Traditional timestamp anomaly flags contributed only marginally to the overall prediction, further underscoring the need for a non-linear, feature-rich classification approach.

2) *Isolation Forest Failure Analysis.* The Isolation Forest model's failure to achieve forensically viable performance is attributed to two factors: contamination parameter sensitivity and feature space overlap. The model's contamination parameter, calibrated to the extremely low observed anomaly rate, resulted in excessive conservatism, leading to the model missing over one hundred out of one hundred and two actual timestomping events.

More fundamentally, analysis of the anomaly score distribution revealed a near-identical mean score between benign and timestomped files, confirming substantial overlap in feature space. This violation of the fundamental assumption of Isolation Forest, that anomalies are isolated faster than benign points, proves that timestomped files are not easily distinguishable by individual feature deviations alone. Consequently, the Isolation Forest is functionally ineffective for forensic triage in this specific context.

3) *Forensic Triage Implications.* From the perspective of a digital forensics practitioner, the model comparison provides clear and distinct operational guidance.

The supervised Random Forest classifier offers an acceptable compromise between detection and efficiency. It successfully identifies the majority of malicious files while simultaneously achieving a massive reduction in the scope of manual review. The resulting number of incorrectly flagged benign files is a manageable burden when weighed against the hundreds of thousands of files an investigator would otherwise be forced to review.

Conversely, the Isolation Forest demonstrated a near-complete failure to detect manipulated files. The model missed almost every single timestomped event, resulting in an unacceptable false negative rate that fundamentally compromises the integrity of any forensic investigation. Therefore, it is entirely unsuitable for operational use in this context.

C. Limitations

The current findings, while establishing a methodological benchmark, are subject to limitations that necessitate further research. The dataset used to train and validate the models is derived exclusively from controlled environments, with manipulations constrained to specific, known anti-forensic tools. Consequently, the model's generalization to other timestomping techniques, such as those employing Windows API calls or specialized PowerShell scripts, remains to be empirically tested. Future work must prioritize expanding the training dataset to include a wider variety of adversarial attack signatures.

Furthermore, while the Random Forest classifier proved significantly superior to the unsupervised baseline, a complete comparative analysis against other high-performance supervised algorithms is warranted. Specifically, the pipeline should be extended to explore and benchmark the performance of powerful models like XGBoost and other Gradient Boosting Machines (GBM) to identify the absolute optimal model for deployment.

Finally, the Random Forest model's achieved scores suggest conservative regularization; therefore, an ongoing effort is

required to explore slightly more aggressive synthetic oversampling ratios and alternative class weighting penalties to optimize the inherent trade-off between Precision and Recall, with the goal of minimizing the number of missed files at an acceptable cost to triage overhead. The final step remains the full integration of the optimal model into the Autopsy Ingest Module for operational validation.

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